Missing values in a control variable

Empirical Methods for Law and Economics | EDLE

Supplementary Material

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The question

- We need to control for **income**. Income is observed for some observations and missing for others.
- By default, Stata drops every observation with a missing value in any variable of the regression (*listwise deletion*, or *complete-case analysis*).
- **Question from class:** can we keep the full sample? Are there techniques to fill in the missing values, and are they commonly used?

Short answer: yes, there are. Which one is appropriate depends less on the technique than on *why* the values are missing.

What we lose by dropping observations

Two different costs, which are easy to confuse:

- **Precision.** Fewer observations, larger standard errors. Always true.
- **Bias.** The complete cases may be a selected subsample. Whether this happens depends on the missingness mechanism.

A useful result: in a regression, complete-case OLS stays unbiased whenever missingness depends only on the regressors (and not on Y given the regressors). Dropping observations is therefore often less harmful than it looks; it is the bad imputations that can do more damage.

Little, R. J. A. and D. B. Rubin (2019), Statistical Analysis with Missing Data, 3rd ed., Wiley.

Why values are missing: three mechanisms (Rubin, 1976)

Mechanism	Missingness depends on...	Income example
MCAR completely at random	nothing: a coin flip	a page of questionnaires lost in the mail
MAR at random	only *observed* variables	retired respondents skip the question more often, and we observe who is retired
MNAR not at random	the *missing value itself*	people with very high (or very low) incomes refuse to answer

- MAR cannot be tested with the data: it is always an assumption to be argued.

DISCUSS Which mechanism?

A household survey asks respondents for their exact annual income. About one in five does not answer.

Which mechanism would you bet on? What could you look at in the data to make your bet more informed?

ANSWER Which mechanism?

- Income nonresponse is the textbook case of **MNAR**: refusal is more frequent at the tails of the distribution, so it depends on income itself.
- MCAR is the least plausible: nonrespondents are almost never a random subsample.
- What we *can* look at:
  - regress a missing indicator on observed characteristics: if it is predictable, MCAR is rejected;
  - check whether the survey asks nonrespondents for an **income bracket**: this turns a large part of the problem into MAR, or even into an interval;
  - collect strong predictors of income (education, occupation, housing tenure, household size, region): the more of income they explain, the more defensible MAR becomes.

The toolbox, from least to most defensible

Method	Main problem
Mean or median imputation	shrinks the variance, weakens correlations
Missing-value dummy	biased in observational data, even under MCAR
Single imputation (regression, hot deck, *k*NN)	treats imputed values as true: standard errors too small
Multiple imputation (MICE)	standard in applied work under MAR
FIML / EM	equivalent to multiple imputation under MAR
Inverse probability weighting	reweights complete cases; common for attrition in panels
Machine learning (missForest, . . .)	good predictions; for inference, use inside multiple imputation

PREDICT Mean imputation

We replace every missing income with the sample mean of income and run the regression on the full sample.

What happens to the variance of income, to its correlation with the other variables, and to the coefficient on income?

ANSWER Mean imputation

- **Variance falls**: all imputed observations sit at one point.
- **Correlations weaken**: the imputed points carry no relation with Y or with the other regressors.
- The **coefficient on income is biased**, typically toward zero; and because income is a worse control, the coefficient on our variable of interest X keeps part of the bias that income was supposed to remove.
- Standard errors are too small: the regression believes it has more information than it does.

Mean imputation is a quick fix for prediction tasks, not for inference.

The missing-value dummy

Replace missing income with 0 and add $D_i = 1$ if income was missing:

$$Y_i = a + \beta X_i + \gamma \widetilde{\text{Inc}}_i + \lambda D_i + e_i$$

- Very common in applied economics, because it is simple and keeps the full sample.
- In observational data it is **biased even under MCAR** (Jones, 1996): for the observations with $D_i = 1$ we are not controlling for income at all.
- **Acceptable exception:** pre-treatment covariates in a randomised experiment. There X is independent of the covariates by design, so the control only serves precision and $\hat{\beta}$ stays unbiased.

Jones, M. P. (1996), Journal of the American Statistical Association.

Multiple imputation: the idea

1. Model income as a function of the other variables, and **draw** plausible values for each missing observation, adding random noise.
2. Repeat m times (e.g. $m = 20$): m complete datasets that differ only in the imputed values.
3. Estimate the regression in each dataset.
4. Combine the results with **Rubin's rules**:

$$\bar{\beta} = \frac{1}{m}\sum_{j=1}^{m}\hat{\beta}_j, \qquad T = \underbrace{\bar{W}}_{\text{average within variance}} + \left(1 + \frac{1}{m}\right)\underbrace{B}_{\text{variance across imputations}}$$

- B is what single imputation forgets: our uncertainty about the missing values. This is why the standard errors come out right.

PREDICT Who goes into the imputation model?

To impute income we use education, occupation and region. A colleague says: "Do not put Y in the imputation model. Using the outcome to build a regressor is circular."

If we follow this advice, what happens to the coefficient on income in the final regression?

ANSWER Who goes into the imputation model?

- The advice is intuitive but **wrong**. If Y is left out, the imputed incomes are drawn as if they had no relation with Y.
- In the final regression, the imputed observations dilute the income–Y relationship: the coefficient on income is **biased toward zero**, and income controls less than it should.
- Including Y is not circular: it only preserves in the imputed data the correlations present in the observed data.
- Two practical rules:
  - include Y and X in the imputation model;
  - the imputation model must be at least as rich as the analysis model (same controls, same interactions, same fixed effects or a coarser version of them).

White, I. R., P. Royston and A. M. Wood (2011), "Multiple imputation using chained equations", Statistics in Medicine.

STATA Multiple imputation of income

```
mi set wide
mi register imputed lninc
mi impute chained (pmm, knn(10)) lninc = y x edu i.occ i.region i.year, ///
    add(20) rseed(12345)
mi estimate: regress y x lninc i.region i.year, vce(cluster id)
```

- `pmm` (predictive mean matching) imputes only values actually observed in the data: no negative incomes, and the skewness of income is preserved.
- `y` and `x` are in the imputation model, as the previous slide requires.
- With high-dimensional fixed effects: `mi estimate, cmdok: reghdfe ...`; in the imputation model, use fixed effects at a coarser level to avoid convergence problems.
- 20 imputations are enough if the share of missing values is below about 20–30%; with more missing values, increase m.
- If the survey records income brackets for nonrespondents, `mi impute intreg` draws income inside the declared interval.

When MAR is not credible

No imputation fixes MNAR on its own. Options:

- **Selection models** (Heckman): need a variable that affects whether income is reported but not income itself. Rarely credible.
- **Bounds**: give up a point estimate and report a range.
  - Lee (2009) bounds: standard for differential attrition in experiments (`leebounds` from SSC).
  - Manski bounds: minimal assumptions, often very wide.
- **Sensitivity analysis** ("delta adjustment"): shift the imputed incomes by $\pm\delta$ to mimic nonrespondents who earn systematically more or less than MAR predicts, and report how large δ must be to overturn the result.

Lee, D. S. (2009), Review of Economic Studies. Manski, C. F. (1989), Journal of Human Resources.

Is multiple imputation common in economics?

- **Standard** in epidemiology, biostatistics and quantitative sociology.
- **Relatively rare** as the main specification in economics journals; more often a robustness check in the appendix. What we see most often:
  - complete cases, plus a table showing that missingness is unrelated to X;
  - the missing-value dummy, especially when X is (quasi-)randomised;
  - administrative or linked data, which avoid the problem at the source.
- Referees accept multiple imputation, provided the complete-case results are shown next to it.

The paradox: you probably already use imputed data

Imputation is mostly done *upstream*, by the data producers:

- **Survey of Consumer Finances** (Federal Reserve) and **HFCS** (ECB): released with five imputed versions ("implicates") of each record.
- **Current Population Survey** (US): missing earnings imputed with a hot deck.
- **EU-SILC**: missing income components imputed by the national statistical offices.

Why it matters: the hot deck does not condition on every variable in *your* regression. For the variables it ignores (e.g. union membership), the imputed observations dilute the relationship and the coefficient is biased toward zero: **match bias** (Bollinger and Hirsch, 2006).

Always look for imputation flags in the documentation.

Bollinger, C. R. and B. T. Hirsch (2006), Journal of Labor Economics.

DISCUSS A referee report

You are refereeing a paper on the effect of a local policy on crime. Household income is a control and is missing for 25% of the sample. The authors use the missing-value dummy in all tables, without further comment.

What would you ask the authors? Write two requests.

ANSWER A referee report

Reasonable requests, in order of cost for the authors:

1. A **balance check** of missingness: regress the missing indicator on the policy variable and pre-treatment characteristics. If missingness is unrelated to the policy, the dummy is a minor issue.
2. The main table **side by side** in three versions: complete cases, missing-value dummy, multiple imputation. Similar estimates close the issue.
3. The results **without income** as a control: if income is not essential to the identification argument, the question loses importance.
4. If the estimates diverge: a sensitivity analysis on departures from MAR, or bounds.

A good request explains *why* it matters for the paper's conclusion; "use multiple imputation" alone is not a good request.

STATA Real data: union membership in nlsw88

```
sysuse nlsw88, clear
generate lnwage = ln(wage)
misstable summarize                          // union is missing for many women
generate byte union_miss = missing(union)
logit union_miss collgrad ttl_exp i.race south smsa married   // is it MCAR?

local ctrl "collgrad ttl_exp i.race south smsa married"
regress lnwage union `ctrl', vce(robust)                       // (1) complete cases
generate union0 = cond(union_miss, 0, union)
regress lnwage union0 union_miss `ctrl', vce(robust)          // (2) missing dummy
mi set wide
mi register imputed union
mi impute logit union = lnwage `ctrl', add(20) rseed(2026)
mi estimate: regress lnwage union `ctrl', vce(robust)          // (3) multiple imputation
```

- Before running: in which specification do you expect the largest sample, and the smallest standard errors on `collgrad`?
- Compare the coefficients on `union` and `collgrad` across (1), (2), (3). Then rerun (3) without `lnwage` in the imputation model.

Take-aways

- First ask *why* the values are missing (MCAR, MAR, MNAR); the technique comes second.
- Mean imputation and single imputation make standard errors too small; the missing dummy is fine in experiments, risky otherwise.
- Under MAR, multiple imputation keeps the full sample with correct standard errors, provided the imputation model includes Y and X.
- In economics, report it next to the complete-case results rather than instead of them.
- Check whether your data are already imputed by the producer.

References

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